CFIMNet: Cross-Modal Frequency-Guided Iterative Multiscale Feature Fusion Network for Multimodal Remote Sensing Object Detection

Jianning Chi, *Member, IEEE*, Feihu Yin, Qingwang Wang, *Senior Member, IEEE*, Geng Lin, Tao Wang, and Xiaoyang Xu

***Abstract*—Multimodal object detection can improve detection accuracy and robustness by fusing complementary information from different modalities. However, in remote sensing scenarios, complex environments often weaken data reliability and cross-modal complementarity, making fixed-pattern fusion strategies insufficient for robust and accurate detection. To address these challenges, this article proposes CFIMNet, a cross-modal frequency-guided iterative multiscale feature fusion (IMFF) network that improves feature complementarity and multiscale representation for RGB–infrared remote sensing object detection. Specifically, a cross-modal frequency enhancement (CFE) module is designed to model channel–spatial interactions between RGB and infrared features and enhance structural information and fine-grained details through adaptive frequency decomposition and reconstruction. In addition, an IMFF module is introduced to progressively fuse cross-modal features across multiple scales within a parameter-shared iterative framework, thereby improving semantic consistency while preserving modality-specific characteristics. Extensive experiments on three benchmark remote sensing datasets demonstrate that CFIMNet consistently outperforms the state-of-the-art methods in detection accuracy and robustness. Code will be available at https://github.com/MapleMoonY/CFIMNet**

***Index Terms*—Cross-modal fusion, frequency enhancement, iterative multiscale fusion, remote sensing object detection.**

I. Introduction

In REMOTE sensing object detection, multimodal fusion has attracted increasing attention because it enables detectors to exploit complementary information acquired by heterogeneous sensors. In particular, the RGB images provide rich texture and appearance cues; however, their reliability may deteriorate under inappropriate exposure settings and varying illumination conditions, thereby degrading downstream detection performance [1]. In contrast, the infrared images are less sensitive to illumination variations and can effectively highlight thermally salient targets [2]. The complementary use of these two modalities therefore enables more reliable perception in challenging scenes characterized by low illumination, cluttered backgrounds, and substantial appearance variations, making RGB–infrared fusion a promising solution for remote sensing applications.

The existing multimodal remote sensing object detection methods can generally be categorized as pixel-level, feature-level, or decision-level fusion. Pixel-level fusion combines raw RGB and infrared images at the input stage and preserves low-level information [3]; however, it is often sensitive to cross-modal discrepancies, spatial misalignment, and sensor noise. Decision-level fusion integrates the predictions of modality-specific detectors and is relatively straightforward to implement [4], but it provides limited cross-modal interaction because the two modalities are combined only at the prediction stage. By contrast, feature-level fusion facilitates cross-modal interaction within the representation space, enabling the network to learn more discriminative and complementary features [5]. Consequently, it has become the dominant paradigm for multimodal remote sensing object detection.

The existing feature-level fusion methods improve multimodal detection from several perspectives, including cross-modal alignment and calibration [6], attention-guided feature interaction [7], [8], consistency- or contrast-driven fusion [9], and hybrid convolution–transformer architectures [10]. Although these methods effectively reduce modality mismatch and strengthen semantic interaction between RGB and infrared features, most operate primarily in the spatial or semantic domain and do not explicitly model frequency-domain complementarity across modalities [11]. Consequently, fine-grained textures, object contours, and small-object boundaries may be attenuated during fusion. Moreover, many multiscale fusion strategies rely on single-pass pyramid aggregation or attention-based reweighting. Although such strategies enhance scale-aware representation, they may not progressively correct cross-modal bias or semantic inconsistency across scales. These limitations motivate the development of a feature-level fusion framework that jointly performs cross-modal frequency enhancement (CFE) and iterative multiscale refinement.

To address these limitations, we propose CFIMNet, a cross-modal frequency-guided iterative multiscale feature fusion (IMFF) network for RGB–infrared remote sensing object detection. CFIMNet improves feature-level fusion from two complementary perspectives: enhancing cross-modal details

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Jianning Chi is with the Faculty of Robot Science and Engineering, Northeastern University, Shenyang 110167, China (e-mail: chijianning@mail.neu.edu.cn).

Feihu Yin, Geng Lin, Tao Wang, and Xiaoyang Xu are with the College of Information Science and Engineering, Northeastern University, Shenyang 110167, China (e-mail: yinfh@mails.neu.edu.cn; ling1@mails.neu.edu.cn; wangt14@mails.neu.edu.cn; xuxy7@mails.neu.edu.cn).

Qingwang Wang is with the Faculty of Information Engineering and Automation, Kunming University of Science and Technology, Kunming 650500, China (e-mail: wangqingwang@kust.edu.cn).

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in the frequency domain and progressively refining multimodal representations across spatial scales. Unlike Fourier- or wavelet-based approaches that decompose features using predefined transforms, the proposed CFE module first performs channel–spatial interaction between the two modalities and then reconstructs modality-specific features through adaptive frequency decomposition and cross-modal frequency weighting. Moreover, instead of employing feature pyramid network (FPN)/path aggregation network (PAN)-style single-pass pyramid aggregation or single-step progressive fusion, the proposed IMFF module repeatedly applies a parameter-shared fusion block to refine RGB and infrared features across scales. By integrating frequency-aware enhancement with iterative multiscale refinement, CFIMNet produces more discriminative and robust multimodal representations for complex remote sensing scenes.

The main contributions of this work are summarized as follows.

1) We propose CFIMNet, a cross-modal frequency-guided IMFF network for RGB–infrared remote sensing object detection. CFIMNet jointly incorporates frequency-guided feature enhancement and progressive multiscale refinement to improve multimodal representation and detection robustness in complex remote sensing scenes.
2) We design a CFE module to strengthen cross-modal complementarity. By integrating channel–spatial interaction with adaptive frequency decomposition and reconstruction, CFE enhances structural information and fine-grained details in modality-specific features before fusion.
3) We develop an IMFF module that progressively refines and integrates RGB and infrared features across multiple spatial scales using shared parameters. IMFF improves cross-scale semantic consistency while preserving modality-specific characteristics, thereby enhancing the representation and detection of small objects and fine-grained structures.

II. Related Work

A. RGB–Infrared Fusion for Remote Sensing Object Detection

RGB–infrared fusion has emerged as an important research direction in remote sensing object detection because it combines the rich appearance cues of RGB imagery with the illumination-robust thermal responses of infrared imagery [12]. The existing methods can be broadly categorized according to their fusion mechanisms. Feature-level approaches interact RGB and infrared representations within the feature space and have become the dominant paradigm because they exploit cross-modal complementarity more effectively than pixel-level or decision-level fusion. Attention-based approaches introduce channel-, spatial-, or modality-aware weighting to emphasize informative responses and suppress unreliable features. Transformer-based methods capture long-range cross-modal dependencies and strengthen semantic interaction in complex scenes. Alignment-aware approaches aim to reduce modality mismatch arising from sensor-specific imaging characteristics and spatial misregistration, thereby improving the reliability of the fused representations.

The representative methods have explored these directions from different perspectives. For example, C^2Former [13] performs calibrated cross-modal interaction to alleviate modality mismatch, whereas SuperYOLO [14] introduces a super-resolution-assisted detection framework to improve small-object representation. Dynamic gating and expert-based fusion [15], cross-modal difference modeling [16], and hierarchical dual-stream interaction [17] have also been investigated to facilitate adaptive fusion and strengthen multimodal feature representation. Collectively, these methods have advanced RGB–infrared object detection by improving spatial alignment, semantic interaction, and adaptive feature aggregation.

However, most existing RGB–infrared fusion methods operate primarily in the spatial or semantic domain, whereas frequency-domain complementarity between RGB and infrared features remains insufficiently explored. Consequently, fine-grained textures, object contours, and small-object boundaries may be attenuated during fusion, particularly in cluttered aerial scenes. Moreover, many existing strategies aggregate multilevel features through fixed or single-stage fusion, which may not adequately preserve modality-specific characteristics when object scales vary substantially. To address these limitations, CFIMNet employs CFE to enhance complementary frequency-domain cues before fusion and uses IMFF to progressively refine cross-modal representations across multiple spatial scales.

B. Frequency-Domain Feature Modeling

The frequency-domain feature modeling has recently attracted increasing attention in multimodal image analysis because it provides an explicit means of separating global structural information from local details [18]. In remote sensing imagery, the low-frequency components generally encode the global scene layout, large-scale background variations, and smooth illumination or thermal distributions. Midfrequency components commonly capture object contours, structural transitions, and local background variations, whereas high-frequency components primarily represent edge textures, small-object boundaries, and fine-grained details. In RGB–infrared remote sensing imagery, RGB features typically contain richer appearance and texture cues, whereas infrared features are less sensitive to illumination variations and more effectively highlight thermally salient targets. Frequency-domain modeling can therefore help distinguish modality-specific responses and strengthen complementary information across the two modalities.

The existing frequency modeling approaches can be broadly divided into predefined transform-based methods and learnable frequency modeling methods. Fourier-based approaches represent features in the global frequency domain and are effective for analyzing spectral responses [19]. Wavelet-based approaches decompose features into multiresolution subbands, enabling the representation of localized structural details [20]. The discrete cosine transform (DCT)-based methods provide compact frequency representations and have been employed to emphasize discriminative components in visual features

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[21]. Beyond these predefined transforms, the learnable frequency modeling methods use adaptive filters or weighting mechanisms to dynamically emphasize informative frequency components, thereby improving boundary awareness and fine-detail representation [22].

Although these methods improve frequency-aware representation, their applicability to RGB–infrared remote sensing object detection remains limited. Transform-based approaches generally rely on predefined bases or fixed frequency partitions, whereas learnable frequency filtering methods often reweight frequency responses within an individual feature stream. Consequently, frequency modeling is commonly treated as an independent enhancement process rather than being tightly integrated with cross-modal interaction. In contrast, the proposed CFE module first performs channel–spatial interaction between RGB and infrared features and subsequently generates frequency reconstruction weights from joint cross-modal statistics. This design not only enhances informative frequency responses but also adaptively selects complementary low-, mid-, and high-frequency cues from the two modalities for object detection.

C. Multiscale Feature Modeling

Multiscale feature modeling is essential for remote sensing object detection because targets in aerial scenes often exhibit substantial variations in scale, while small objects contain limited but critical structural information. The existing approaches can be broadly categorized into feature-pyramid architectures, attention-based multiscale fusion methods, and progressive or iterative refinement methods. The FPN- and PAN-style architectures enhance cross-scale representation through top–down, bottom–up, or bidirectional feature propagation. However, feature fusion at each scale is generally performed only once, which may attenuate fine-grained details [23], [24], [25]. The attention-based multiscale methods assign adaptive weights to different scales or spatial regions, thereby emphasizing informative features in complex backgrounds. Progressive fusion methods further aggregate features in a stagewise manner to strengthen contextual representation.

Recent studies have investigated these directions in remote sensing object detection. For example, transformer-based multiscale fusion has been employed to strengthen contextual interaction [26], while the combination of multiscale fusion and knowledge distillation has improved the detection of small and occluded aerial targets [27]. Progressive fusion and attention-based adaptive weighting have also been introduced to enhance feature aggregation [28], [29], whereas lightweight multireceptive-field enhancement has been adopted to improve small-target representation [30]. Although these methods improve multiscale representation from different perspectives, they generally rely on pyramid-based propagation, direct feature reweighting, or stagewise aggregation.

In complex RGB–infrared scenes, however, these strategies may not simultaneously preserve cross-scale semantic consistency and modality-specific details. The FPN- and PAN-style architectures primarily propagate features along predefined paths, whereas attention-based methods usually perform direct feature reweighting without explicitly refining the two modality streams. In contrast, the proposed IMFF module repeatedly applies a shared multiscale fusion block through a parameter-shared iterative update scheme. At each iteration, the scale-aware and position-aware gates generate a fused representation, toward which the RGB and infrared features are progressively refined. This design mitigates cross-modal semantic bias and preserves complementary modality-specific information while introducing only a limited number of additional parameters.

III. Proposed Method

A. Architecture Overview

The overall framework of CFIMNet is shown in Fig. 1. Built upon a Siamese backbone for RGB and infrared feature extraction, CFIMNet mainly includes two core modules: the CFE module and the IMFF module. The CFE module is responsible for enhancing cross-modal complementary representations before fusion. It performs channel–spatial interaction between RGB and infrared features and further strengthens structural information and fine-grained details through adaptive frequency decomposition and reconstruction. Based on the enhanced features, the IMFF module progressively aggregates cross-modal information across multiple spatial scales within a parameter-shared iterative framework, thereby improving cross-scale semantic consistency while preserving modality-specific characteristics. In this way, CFE focuses on cross-modal feature enhancement, and IMFF further performs progressive multiscale fusion on the enhanced representations. The resulting fused features are finally sent to the detection head for target localization and classification.

B. CFE Module

We propose a CFE module, which enables cross-modal interaction at both the channel and spatial levels while performing adaptive frequency-domain decomposition and reconstruction, thereby enhancing the discriminative capability of the modality-specific features before fusion. It mainly consists of two core components: a cross-modal attention interaction unit (CAIU) and an adaptive frequency decomposition unit (AFDU).

1) Cross-Modal Attention Interaction Unit: As shown in Fig. 1, the CAIU is used to enhance cross-modal feature interaction before frequency decomposition. Given the RGB and infrared feature $F_{rgb}, F_{ir} \in \mathbb{R}^{C\times H\times W}$, the two modalities are first concatenated along the channel dimension to form a joint representation $F_{cat} \in \mathbb{R}^{2C\times H\times W}$. A depthwise separable convolution [31] is then applied to extract cross-modal interaction features $F_{cross} \in \mathbb{R}^{2C\times H\times W}$

$$F_{cat} = \mathrm{Cat}\left(F_{rgb}, F_{ir}\right) \tag{1}$$

$$F_{cross} = \mathrm{DSConv}_{3\times 3}\left(F_{cat}\right) \tag{2}$$

where $Cat(\cdot)$ denotes channel concatenation and $DSConv$ represents depthwise separable convolution.

Based on F_{cross}, channel attention and spatial attention are further computed to modulate the two modalities. Specifically, two channel attention branches are applied to F_{cross},

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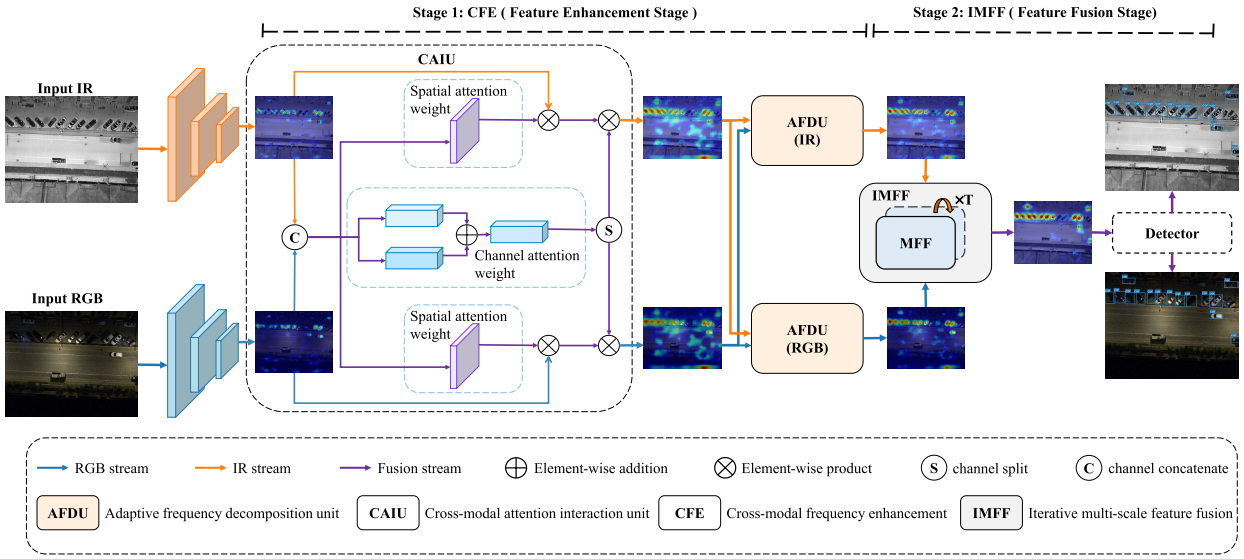


Fig. 1. Overall architecture of CFIMNet. A Siamese backbone extracts multiscale features from RGB and infrared inputs; Stage 1 applies the CFE module, where CAIU performs channel–spatial cross-modal interaction and AFDU conducts adaptive frequency decomposition/reconstruction to enhance modality features; Stage 2 adopts the IMFF module to iteratively refine and fuse the enhanced cross-modal features (with shared parameters), and the final fused features are fed into the detector head for prediction.

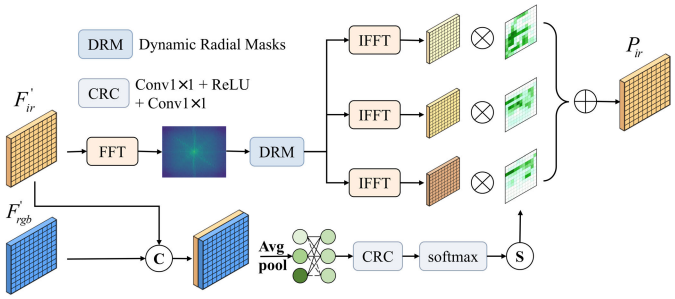


Fig. 2. Architecture of the AFDU illustrated on the infrared branch. The enhanced feature is transformed into the frequency domain by FFT and decomposed using dynamically generated radial masks. The resulting low-, mid-, and high-frequency components are transformed back by IFFT and adaptively reconstructed using cross-modal frequency weights.

and their outputs are averaged to obtain the channel weight $w_{ch} \in \mathbb{R}^{2C\times1\times1}$. In parallel, a spatial attention branch is used to generate the spatial weight $w_{sp} \in \mathbb{R}^{1\times H\times W}$

$$w_{ch} = \frac{1}{2}\left(\phi_{ch}^{low}\left(F_{\text{cross}}\right) + \phi_{ch}^{high}\left(F_{\text{cross}}\right)\right) \tag{3}$$

$$w_{sp} = \phi_{sp}\left(F_{\text{cross}}\right) \tag{4}$$

where $\phi_{ch}^{low}(\cdot)$ and $\phi_{ch}^{high}(\cdot)$ denote two channel-attention mapping functions and $\phi_{sp}(\cdot)$ denotes the spatial-attention mapping function.

The channel weight w_{ch} is then split into $w_{ch}^{rgb} \in \mathbb{R}^{C\times1\times1}$ and $w_{ch}^{ir} \in \mathbb{R}^{C\times1\times1}$ for the RGB and infrared branches, respectively. Finally, the enhanced RGB feature $F'_{rgb} \in \mathbb{R}^{C\times H\times W}$ and infrared feature $F'_{ir} \in \mathbb{R}^{C\times H\times W}$ are obtained by jointly applying the channel weights and spatial weights through weighted modulation, as formulated in the following:

$$F'_{rgb} = F_{rgb} \otimes w_{ch}^{rgb} \otimes w_{sp} \tag{5}$$

$$F'_{ir} = F_{ir} \otimes w_{ch}^{ir} \otimes w_{sp} \tag{6}$$

where $\otimes$ denotes the elementwise multiplication.

2) Adaptive Frequency Decomposition Unit: To further capture frequency-domain complementarity between the two modalities, the AFDU performs explicit frequency-domain decomposition on the enhanced features and reconstructs them adaptively with learnable weights. Taking the infrared feature as an example, the overall structure of AFDU is illustrated in Fig. 2. Specifically, for the input feature F'_{ir}, its frequency-domain representation can be expressed as

$$\hat{F}'_{ir} = \mathcal{F}\left(F'_{ir}\right) \tag{7}$$

where $\mathcal{F}(\cdot)$ denotes the 2-D Fourier transform.

Based on the current feature-map size, three radial frequency masks are dynamically generated in the frequency domain, denoted by M_{low}, M_{mid}, and M_{high}, respectively. Let (ξ, η) denote the discrete frequency coordinate of an arbitrary frequency location. Its normalized radial distance can be defined as

$$\tilde{r}\left(\xi,\eta\right) = \frac{\sqrt{\xi^2+\eta^2}}{r_{\max}} \tag{8}$$

where $r_{\max}$ denotes the maximum radial distance in the current frequency plane. Based on $\tilde{r}(\xi,\eta)$, the unnormalized response of the kth frequency band is modeled by a Gaussian radial function

$$\mathcal{K} = \{\text{low}, \text{mid}, \text{high}\} \tag{9}$$

$$\tilde{M}_k\left(\xi,\eta\right) = \exp\left(-\frac{\left(\tilde{r}\left(\xi,\eta\right)-\mu_k\right)^2}{2\sigma_k^2}\right), \quad k \in \mathcal{K} \tag{10}$$

where μ_k and σ_k denote the center frequency and bandwidth of the kth radial frequency band, respectively. This formulation provides a smooth frequency response for each band and avoids abrupt transitions between adjacent frequency regions. The final frequency masks are then normalized at each frequency location as

$$M_k\left(\xi,\eta\right) = \frac{\tilde{M}_k\left(\xi,\eta\right)}{\sum_{j\in\mathcal{K}}\tilde{M}_j\left(\xi,\eta\right)+\varepsilon}, \quad k \in \mathcal{K} \tag{11}$$

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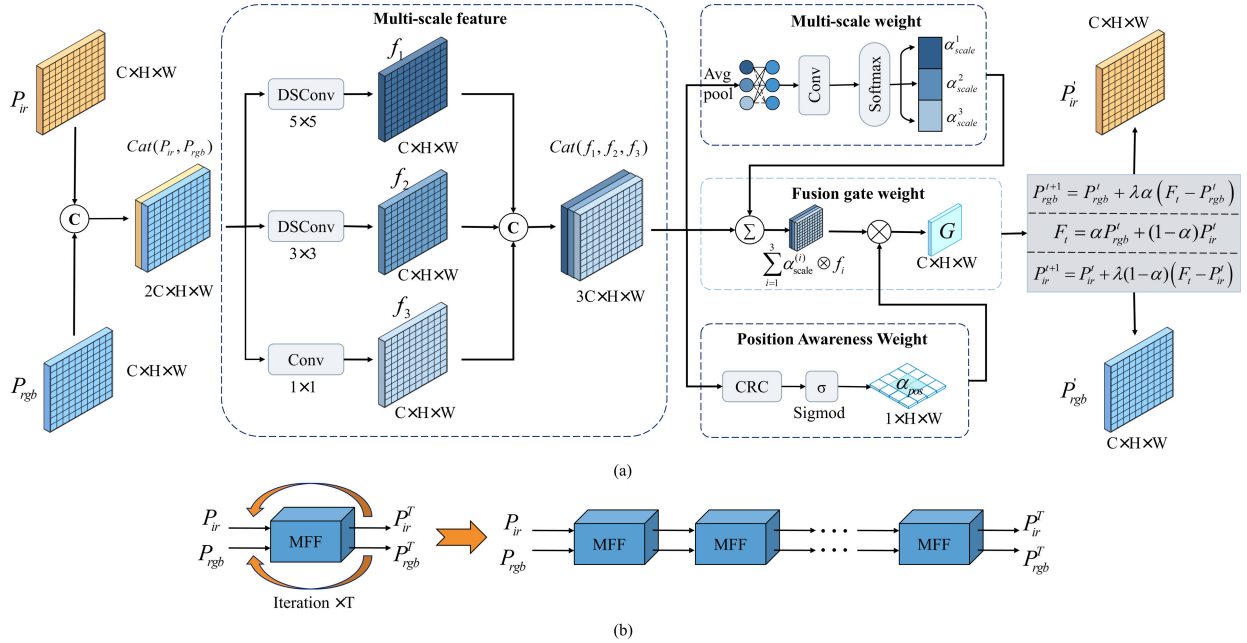


Fig. 3. Architecture of the IMFF module. (a) One IMFF block: concatenated RGB/infrared features are processed by multiscale convolutional branches, and scale-aware weights together with a position-aware branch generate a fusion gate to update and refine modality-specific features. (b) Parameter-shared iterative refinement, where the same block is applied for T iterations (an unrolled view is shown).

where ε is a small constant to avoid division by zero. Therefore, the following constraint holds:

$$\sum_{k\in\mathcal{K}} M_k(\xi,\eta) \approx 1. \tag{12}$$

Based on the above masks, the low-, mid-, and high-frequency components of F'_{ir} can be obtained as

$$x_{ir}^{k} = \mathcal{F}^{-1}\left(\hat{F}'_{ir} \otimes M_k\right), \quad k \in \mathcal{K} \tag{13}$$

where $\mathcal{F}^{-1}(\cdot)$ denotes the 2-D inverse Fourier transform.

Subsequently, the two-modal enhanced features are concatenated to form the concatenated feature $F'_{cat} = \mathrm{Cat}(F'_{rgb}, F'_{ir}) \in \mathbb{R}^{2C\times H\times W}$. Then, the concatenated feature F'_{cat} is fed into a global average pooling (GAP) layer to extract global statistical information. After CRC mapping, the output is normalized by a Softmax function to obtain three sets of frequency weights

$$\left(w_{\text{low}}, w_{\text{mid}}, w_{\text{high}}\right) = \mathrm{Softmax}\left(\mathrm{CRC}\left(\mathrm{GAP}\left(F'_{cat}\right)\right)\right) \tag{14}$$

where $w_{\text{low}}, w_{\text{mid}}, w_{\text{high}} \in \mathbb{R}^{C\times 1\times 1}$ represent the low-frequency, midfrequency, and high-frequency weights, respectively. $CRC(\cdot)$ denotes a sequence consisting of a 1×1 convolution, a ReLU activation function, and another 1×1 convolution, while GAP indicates GAP.

These weights are then combined with the corresponding frequency components to perform weighted reconstruction of the infrared features, yielding the final enhanced infrared feature $P_{ir} \in \mathbb{R}^{C\times H\times W}$

$$P_{ir} = \sum_{k\in\mathcal{K}} w_k \otimes x_{ir}^{k}. \tag{15}$$

The same decomposition and reconstruction procedure is symmetrically applied to the RGB branch, producing the enhanced RGB feature $P_{rgb} \in \mathbb{R}^{C\times H\times W}$.

C. Iterative Multiscale Feature Fusion

To fully exploit the complementary feature relationships between RGB and infrared modalities at different spatial scales and mitigate semantic bias and information loss caused by single-pass fusion, we propose an IMFF module. This module employs an iterative refinement mechanism with shared parameters to achieve gradual, fine-grained fusion of cross-modal features across multiple scales and spatial dimensions while retaining modality-specific characteristics. Its specific architecture is illustrated in Fig. 3. In the first iteration, given an input feature $U = \mathrm{Cat}(P_{rgb}, P_{ir}) \in \mathbb{R}^{2C\times H\times W}$, let $S = \{1, 3, 5\}$ denote the set of convolutional kernel sizes. The multiscale features are extracted as

$$f_k = \mathcal{H}_k(U), \qquad k \in S \tag{16}$$

where $\mathcal{H}_k(\cdot)$ denotes a convolutional operator with a $k\times k$ kernel, which is used to capture contextual information across multiple spatial scales. Then, the multiscale features are concatenated along the channel dimension, and a scale weight vector $\alpha_{\text{scale}} \in \mathbb{R}^{3\times 1\times 1}$ is generated through GAP and Softmax normalization

$$\alpha_{\text{scale}} = \mathrm{Softmax}\left(\mathrm{Conv}_{1\times 1}\left(\mathrm{GAP}\left(\mathrm{Cat}\left(f_1, f_3, f_5\right)\right)\right)\right). \tag{17}$$

Next, a position-awareness mechanism is further introduced to model the importance of different spatial locations. This process generates spatial weights $\alpha_{\text{pos}} \in \mathbb{R}^{1\times H\times W}$ through a lightweight convolutional network

$$\alpha_{\text{pos}} = \sigma\left(\mathrm{CRC}\left(U\right)\right) \tag{18}$$

where σ represents the sigmoid function. Finally, combine the scale attention weight α_{scale} and the position-aware

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weight α_{pos} to generate the cross-modal fusion gating weight $G \in \mathbb{R}^{C\times H\times W}$

$$G = \left(\sum_{k\in S} \alpha_{\text{scale}}^{(k)} \otimes f_k\right) \otimes \alpha_{\text{pos}} \tag{19}$$

where $\alpha_{\text{scale}}^{(k)}$ denotes the scale-aware weight corresponding to the $k \times k$ convolutional branch.

Unlike traditional single-step fusion, the IMFF adopts a parameter-sharing iterative update strategy. At the tth iteration, the current fused feature is first defined as

$$F_t = G^t \otimes P_t^{rgb} + \left(1 - G^t\right) \otimes P_t^{ir}. \tag{20}$$

Based on $F_t \in \mathbb{R}^{C\times H\times W}$, the two modal features are then updated as

$$P_{t+1}^{rgb} = P_t^{rgb} + \lambda G^t \otimes \left(F_t - P_t^{rgb}\right) \tag{21}$$

$$P_{t+1}^{ir} = P_t^{ir} + \lambda \left(1 - G^t\right) \otimes \left(F_t - P_t^{ir}\right) \tag{22}$$

where λ is a fusion coefficient controlling the update magnitude. After T iterations, the final fused feature is obtained

$$F_{out} = G^T \otimes P_T^{rgb} + \left(1 - G^T\right) \otimes P_T^{ir} \tag{23}$$

where $F_{out} \in \mathbb{R}^{C\times H\times W}$ is the final fused feature.

IV. Experiments

A. Datasets

We conducted experiments on three representative RGB–infrared object detection datasets, namely, DroneVehicle [32], VEDAI [33], and FLIR [34], to evaluate the detection accuracy and robustness of CFIMNet.

1) *DroneVehicle:* The DroneVehicle dataset primarily consists of registered RGB–infrared image pairs captured from an uncrewed aerial vehicle (UAV) perspective for ground-vehicle detection. It covers diverse scenes, including urban roads, residential areas, and parking lots, under illumination conditions ranging from daytime to nighttime. The dataset contains five object categories: car, truck, bus, van, and freight car. It provides 19 459 RGB–infrared image pairs, of which 17 990 are used for training and 1469 for testing.
2) *VEDAI:* VEDAI is an open-source aerial-image dataset for small-object detection. It comprises 1268 RGB–infrared image pairs and more than 3700 annotated object instances. The dataset contains eight categories: car, pickup, truck, camping vehicle, boat, tractor, van, and other. For consistent network input, all images are resized to 1024×1024 during both training and testing. The original corner-based annotations are converted into horizontal bounding boxes following [35].
3) *FLIR:* FLIR is a challenging multispectral object detection dataset containing both daytime and nighttime scenes. The aligned version used in our experiments comprises 5142 registered RGB–infrared image pairs, including 4129 training pairs and 1013 test pairs. It contains three object categories: person, vehicle, and bicycle. Because the original image pairs exhibit spatial misalignment, we use the aligned FLIR version provided in [36] for all experiments.

TABLE I
Ablation Study Results of CFIMNet With Different Modules on the DroneVehicle Dataset

Method	CFE	IMFF	mAP50	mAP
Baseline			77.9	52.7
Baseline + CFE	✓		84.3	58.2
Baseline + IMFF		✓	84.6	57.9
CFIMNet (Ours)	✓	✓	86.7	63.2

B. Implementation Details

CFIMNet is implemented in PyTorch and trained and evaluated on a server equipped with an NVIDIA GeForce RTX 3090 GPU. YOLOv5 is adopted as the baseline detector, with CSPDarkNet53 serving as the backbone for multimodal feature extraction. A concatenation-based dual-stream configuration is used as the baseline fusion scheme to facilitate comparison with representative RGB–infrared detection methods built upon the same detection framework. For DroneVehicle and FLIR, the input resolution is set to 640×640, and each model is trained for 100 epochs. For VEDAI, the input resolution is set to 1024×1024, and training is performed for 150 epochs. Stochastic gradient descent (SGD) is used as the optimizer with a momentum of 0.937. A cosine annealing schedule is adopted for learning rate decay, where the initial learning rate is set to 0.01, the final learning rate is set to 0.003, and the weight decay is set to 5×10^{-4}. Detection performance is evaluated using mAP50 (IoU = 0.5), mAP75 (IoU = 0.75), and mAP (IoU = 0.5:0.95). Inference complexity is measured by GFLOPs, and model size is quantified by the number of parameters (Params).

C. Ablation Study

We conduct ablation and robustness experiments on the DroneVehicle dataset to evaluate the contributions of the proposed components. The experiments examine five aspects: the individual and joint effects of CFE and IMFF, the contributions of CAIU and AFDU within CFE, the frequency-mask generation strategy, the number of IMFF iterations, and robustness under image degradation. Unless otherwise specified, all model variants are trained and evaluated using identical settings.

1) Effectiveness of the Key Modules: To assess the contributions of CFE and IMFF, we individually and jointly incorporate the two modules into the baseline. As reported in Table I, the concatenation-based baseline obtains 77.9 mAP50 and 52.7 mAP. Adding CFE increases the results to 84.3 mAP50 and 58.2 mAP, corresponding to absolute improvements of 6.4 and 5.5 points, respectively. Adding IMFF alone yields 84.6 mAP50 and 57.9 mAP, demonstrating the effectiveness of iterative multiscale refinement.

Combining CFE and IMFF produces the best results, reaching 86.7 mAP50 and 63.2 mAP. These results demonstrate that CFE and IMFF provide complementary improvements: CFE

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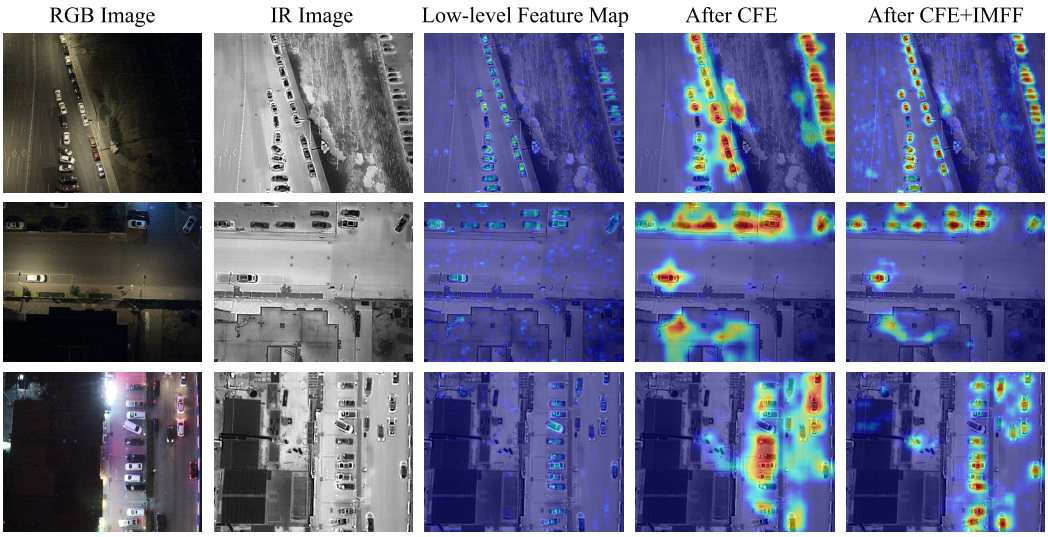


Fig. 4. Effectiveness of CFE and IMFF on feature extraction. Brighter colors highlight the areas of greater model attention.

TABLE II

EFFECTIVENESS OF THE TWO COMPONENTS IN THE CFE MODULE ON THE DRONEVEHICLE DATASET

Method	CAIU	AFDU	mAP50	mAP
Baseline			77.9	52.7
Baseline + CAIU	✓		82.5	55.7
Baseline + AFDU		✓	83.1	55.3
Baseline + CFE	✓	✓	84.3	58.2

TABLE III

EFFECTIVENESS OF THE MASK GENERATION STRATEGY ON THE DRONEVEHICLE DATASET

Masking method	mAP50	mAP	GFLOPs
Hard masks	85.2	61.1	128.6
Smooth masks	85.8	60.5	129.4
Normalized smooth masks	86.7	63.2	129.4

TABLE IV

EFFECTIVENESS OF THE NUMBER OF ITERATIONS IN THE IMFF MODULE ON THE DRONEVEHICLE DATASET

Iterations	mAP50	mAP	Params (M)	GFLOPs
1	85.4	62.1	110.6	119.1
2	86.7	63.2	110.6	129.4
3	85.0	62.8	110.6	139.7
4	84.7	61.6	110.6	149.5
5	84.1	61.0	110.6	159.8

enhances cross-modal structural and frequency information, whereas IMFF progressively refines feature interactions across spatial scales.

The feature visualizations in Fig. 4 provide qualitative support for the ablation results. The low-level feature maps exhibit relatively weak target responses and substantial background activation. After CFE is introduced, the responses around target regions become stronger and more concentrated. With the further addition of IMFF, the activations become more spatially compact, while background responses are reduced. These observations are consistent with the quantitative results in Table I, suggesting that CFE strengthens target-relevant cross-modal features and that IMFF further improves their spatial consistency through iterative multiscale refinement.

2) Effectiveness of the Two Components in the CFE Module: To evaluate the contributions of the two components within CFE, we separately incorporate CAIU and AFDU into the baseline. As shown in Table II, CAIU improves the mAP50/mAP results from 77.9/52.7 to 82.5/55.7, whereas AFDU yields 83.1/55.3. These improvements indicate that both cross-modal attention interaction and adaptive frequency decomposition contribute to feature enhancement. When CAIU and AFDU are jointly incorporated, the complete CFE module achieves 84.3 mAP50 and 58.2 mAP, outperforming both single-component variants. The results therefore demonstrate the complementary effects of cross-modal interaction and frequency-aware reconstruction.

3) Effectiveness of the Mask Generation Strategy: To evaluate the mask generation strategy in AFDU, we compare three designs under identical experimental settings: hard masks, smooth masks, and normalized smooth masks. As shown in Table III, normalized smooth masks achieve the best performance, reaching 86.7 mAP50 and 63.2 mAP. Hard masks introduce abrupt transitions between adjacent frequency bands, whereas smooth masks provide continuous frequency responses. However, without normalization, overlapping responses may lead to inconsistent contributions across frequency bands. The proposed normalization constrains the relative response of each frequency band and provides a more coherent frequency partition.

Compared with smooth masks, normalized smooth masks improve mAP by 2.7 points while retaining the same computational complexity of 129.4 GFLOPs. This result indicates that the performance improvement is associated with the mask normalization strategy rather than an increase in computational cost.

4) Effectiveness of the Number of Iterations in the IMFF Module: To determine the appropriate number of IMFF

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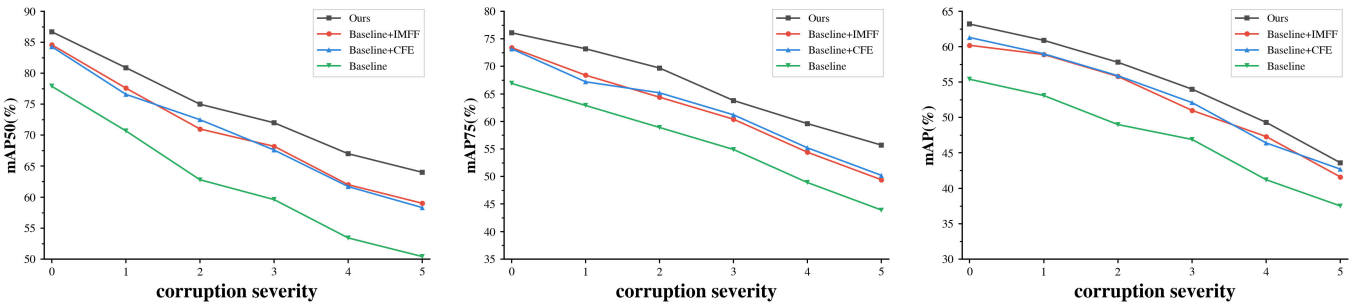


Fig. 5. Robustness evaluation results of different module configurations under image degradation on the DroneVehicle dataset.

TABLE V

COMPARISON OF CFIMNET WITH THE STATE-OF-THE-ART METHODS ON THE DRONEVEHICLE DATASET. "RGB" AND "IR" REPRESENT THE DETECTION RESULTS OF VISIBLE IMAGES AND INFRARED IMAGES, RESPECTIVELY. "RGB + IR" MEANS THE RESULTS OF FUSION DETECTION OF INFRARED AND VISIBLE IMAGES. NUMBERS IN RED INDICATE THE BEST PERFORMANCE, WHILE BOLD NUMBERS REPRESENT THE SECOND-BEST PERFORMANCE

Method	Modality	car	truck	bus	van	freight car	mAP50
Faster RCNN [37]	RGB	82.5	54.1	80.1	41.2	40.1	59.6
R3DET [38]	RGB	79.3	42.2	76.0	28.5	24.5	50.1
ReDet [39]	RGB	69.4	47.8	77.3	29.0	31.4	51.0
YOLOv5s [40]	RGB	89.7	62.0	86.4	46.8	48.8	66.7
PCAF-N [41]	RGB	95.0	65.4	94.5	58.7	50.6	72.8
FFCA-YOLO [42]	RGB	92.1	69.6	93.7	60.8	56.5	74.5
Faster RCNN [37]	IR	89.0	65.0	83.9	42.1	53.1	66.6
R3DET [38]	IR	89.5	48.3	87.1	39.9	16.6	62.3
ReDet [39]	IR	89.4	53.9	79.8	36.5	42.8	60.5
YOLOv5s [40]	IR	94.6	73.1	89.8	55.4	63.4	75.0
PCAF-N [41]	IR	98.0	70.9	96.0	58.9	67.8	78.4
FFCA-YOLO [42]	IR	**98.2**	77.9	**96.9**	66.1	68.8	81.6
UA-CMDet [32]	RGB+IR	87.5	60.7	87.0	37.9	46.8	64.0
ICAFusion [43]	RGB+IR	96.8	74.4	95.3	49.5	61.1	75.4
TSFADet [44]	RGB+IR	89.9	67.9	88.1	54.0	63.7	73.1
C^2Former [13]	RGB+IR	90.2	68.3	89.8	58.5	64.4	74.2
OAFA [45]	RGB+IR	90.3	76.8	90.3	66.0	73.3	79.4
GM-DETR [46]	RGB+IR	92.4	75.3	80.8	90.8	64.9	80.8
SuperYOLO [14]	RGB+IR	97.7	79.0	66.3	**96.6**	67.6	81.4
GHOST [47]	RGB+IR	97.3	78.8	68.9	96.3	66.5	81.5
CMADet [48]	RGB+IR	**98.2**	70.4	78.3	96.8	66.4	82.0
MMFDet [16]	RGB+IR	80.7	97.2	66.7	69.3	98.4	**82.5**
CFIMNet (Ours)	RGB+IR	98.7	**85.1**	97.5	71.1	**81.4**	86.7

iterations, we vary only the iteration count while keeping all other settings unchanged. As reported in Table IV, increasing the number of iterations from one to two improves mAP50 from 85.4 to 86.7 and mAP from 62.1 to 63.2. Further increasing the number of iterations reduces detection accuracy while monotonically increasing GFLOPs. The parameter count remains unchanged because the fusion block shares parameters across iterations.

The performance degradation observed with more than two iterations may be associated with excessive feature smoothing or redundant cross-modal refinement, which can weaken fine-grained discriminative responses. Therefore, two iterations are adopted as the default configuration to balance detection accuracy and computational cost.

5) Robustness Under Image Degradation: Following the corruption protocol in [49], we apply four types of image degradation—fog, frost, motion blur, and snow—to the DroneVehicle test images. Each corruption type is generated at five severity levels. All model variants are trained on the original clean training set using identical settings and are directly evaluated on the corrupted test images without fine-tuning or retraining.

We compare the baseline, Baseline + CFE, Baseline + IMFF, and the complete CFIMNet under increasing corruption severity. As shown in Fig. 5, the performance of all variants decreases as the severity level increases. Nevertheless, CFIMNet consistently achieves the highest mAP50, mAP75, and mAP across the evaluated severity levels. Both single-module

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TABLE VI

COMPARISON OF CFIMNET WITH THE STATE-OF-THE-ART METHODS ON THE VEDAI DATASET. "RGB" AND "IR" REPRESENT THE DETECTION RESULTS OF VISIBLE IMAGES AND INFRARED IMAGES, RESPECTIVELY. "RGB + IR" MEANS THE RESULTS OF FUSION DETECTION OF INFRARED AND VISIBLE IMAGES. NUMBERS IN RED INDICATE THE BEST PERFORMANCE, WHILE BOLD NUMBERS REPRESENT THE SECOND-BEST PERFORMANCE

Method	Modality	car	pickup	camping	truck	other	tractor	boat	van	mAP50
YOLOrs [50]	RGB	85.2	72.9	70.3	50.6	42.6	76.7	18.6	38.9	56.9
YOLOv5s [40]	RGB	83.0	78.9	68.4	74.7	51.7	75.6	33.0	57.8	65.4
FFCA-YOLO [42]	RGB	89.3	86.7	76.9	74.7	51.0	78.5	49.7	74.6	71.7
Faster R-CNN [37]	RGB	80.9	81.6	83.0	76.2	58.0	55.8	69.6	61.5	70.8
SuperYOLO [14]	RGB	90.3	82.6	76.6	68.5	53.8	79.4	58.0	70.3	72.4
YOLOrs [50]	IR	82.0	73.9	63.8	54.2	43.9	54.3	21.9	43.3	54.6
YOLOv5s [40]	IR	81.1	72.8	76.4	66.4	31.9	28.3	51.7	74.7	60.4
FFCA-YOLO [42]	IR	85.5	86.3	81.4	74.8	29.8	68.1	59.9	**96.2**	68.6
Faster R-CNN [37]	IR	85.2	80.5	82.3	80.7	36.1	51.8	56.3	69.8	67.9
SuperYOLO [14]	IR	87.9	81.3	76.9	61.5	39.3	60.5	46.0	71.0	65.6
YOLOrs [50]	RGB+IR	84.1	78.2	68.8	52.6	46.7	67.8	21.4	57.9	59.7
ICAFusion [43]	RGB+IR	84.4	82.5	73.4	76.2	69.5	59.9	53.0	81.9	72.6
CFT [51]	RGB+IR	93.6	83.3	73.9	74.8	51.0	75.9	64.7	72.1	73.7
SuperYOLO [14]	RGB+IR	91.1	85.6	79.3	70.1	53.8	**80.4**	60.2	76.5	75.1
C^2Former [13]	RGB+IR	87.2	80.7	82.7	77.4	59.4	72.9	71.4	75.2	75.7
MROD-YOLO [52]	RGB+IR	**94.0**	**89.7**	75.0	74.8	**73.2**	72.1	68.0	76.2	77.9
ACDF-YOLO [53]	RGB+IR	89.8	87.9	78.5	75.3	67.3	76.2	73.6	75.9	78.1
C^2DFF-Net [54]	RGB+IR	91.8	85.9	**87.2**	74.4	58.7	78.6	62.2	91.3	79.8
MMYFNet [55]	RGB+IR	90.8	88.3	74.5	**84.9**	72.6	74.9	60.0	74.6	80.0
LF-MDet [56]	RGB+IR	92.1	87.3	85.9	71.6	67.1	89.5	68.8	84.1	80.8
ADMPF [57]	RGB+IR	94.4	90.2	84.7	74.4	57.1	74.7	85.4	91.8	**81.6**
CFIMNet (Ours)	RGB+IR	91.5	89.0	90.2	86.4	85.9	59.5	**83.9**	99.5	85.7

variants also outperform the baseline, indicating that CFE and IMFF individually improve robustness to image corruption. The complete model maintains a consistent advantage over the single-module variants, with a more pronounced margin at higher severity levels, demonstrating the complementary contributions of the two modules under degraded observations.

D. Comparison With the State-of-the-Art Methods

To comprehensively evaluate the overall performance of the proposed CFIMNet for multimodal remote sensing object detection, we compare it with several state-of-the-art methods on three representative datasets: DroneVehicle, VEDAI, and FLIR. The compared methods include visible-only detectors, infrared-only detectors, and RGB–infrared fusion detectors. All comparison methods are implemented using either the standardized training protocol described in Section IV-B or their best publicly documented results to ensure fairness and consistency.

1) Quantitative Evaluation: On the DroneVehicle dataset, the quantitative comparison results are reported in Table V. CFIMNet achieves the best overall performance with 86.7 mAP50, outperforming all compared RGB-only, IR-only, and RGB+IR fusion methods. In particular, compared with strong multimodal baselines, such as CMADet (82.0 mAP50) and MMFDet (82.5 mAP50), our method achieves clear gains, demonstrating the effectiveness of the proposed fusion strategy in complex aerial scenes. CFIMNet also obtains competitive categorywise performance, including 98.7 for car, 97.5 for bus, and 85.1 for truck, demonstrating effective categorywise detection performance across different vehicle classes. These results suggest that the proposed CFE and IMFF modules can better exploit complementary RGB–infrared information and improve the representation of structural details and small objects under challenging conditions, such as cluttered backgrounds and illumination variations.

On the VEDAI dataset, the results in Table VI further demonstrate the superiority of CFIMNet. Our method achieves 85.7 mAP50, which is the best performance among all compared methods and significantly surpasses both single-modality detectors and recent multimodal fusion models. Specifically, CFIMNet achieves strong detection accuracy for several representative categories, including 91.5 for car, 89.0 for pickup, 90.2 for camping vehicle, 86.4 for truck, 85.9 for other, 83.9 for boat, and 99.5 for van. Compared with recent multimodal detectors, such as MROD-YOLO (77.9 mAP50), LF-MDet (80.8 mAP50), and ADMPF (81.6 mAP50), CFIMNet shows a more pronounced overall advantage. This indicates that the proposed method is more capable of preserving discriminative details while progressively integrating cross-modal contextual information, thereby improving detection robustness for small targets with

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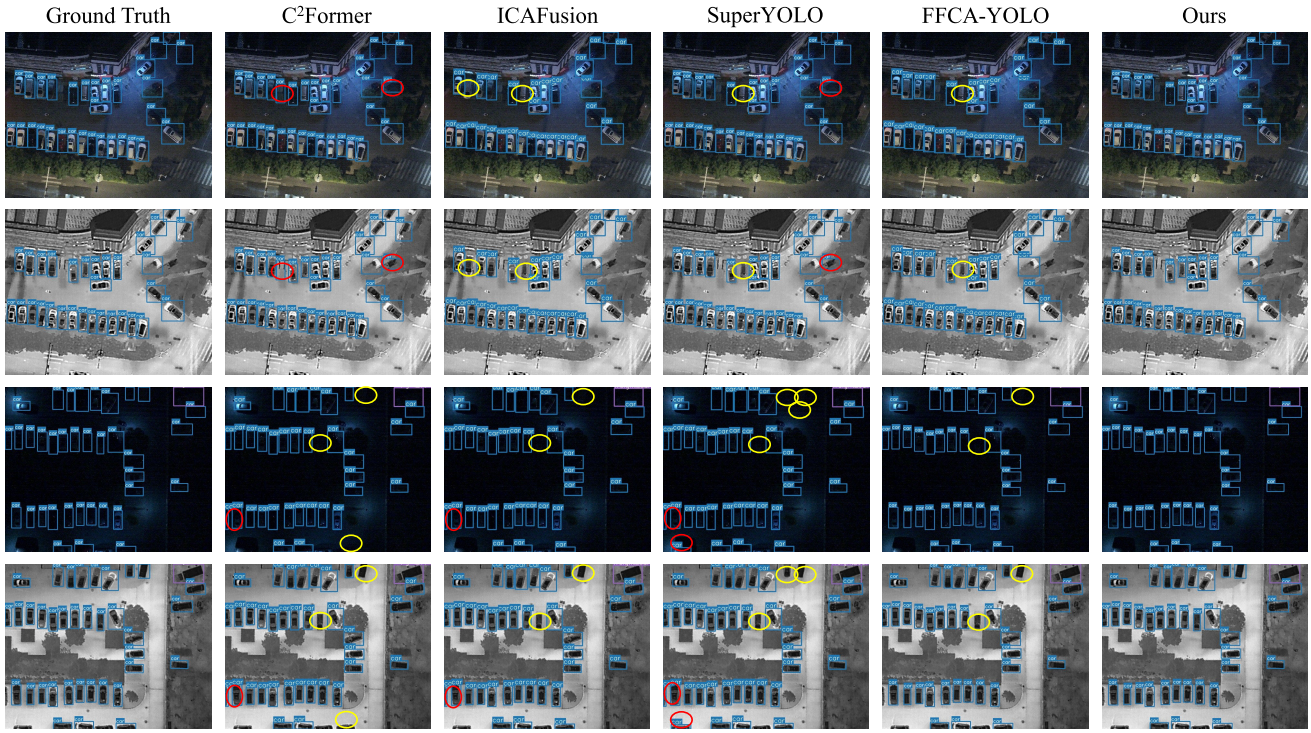


Fig. 6. Qualitative comparison on the DroneVehicle dataset. Red circles indicate false-positive detections and yellow circles indicate missed detections.

TABLE VII
COMPARISON OF CFIMNET WITH THE STATE-OF-THE-ART METHODS ON THE FLIR DATASET

Method	Modality	mAP50	mAP
Faster R-CNN [37]	RGB	62.8	26.4
YOLOv5s [40]	RGB	65.5	24.9
Faster R-CNN [37]	IR	71.9	36.2
YOLOv5s [40]	IR	73.7	35.7
CFT [51]	RGB+IR	78.7	40.2
YOLO Fusion [35]	RGB+IR	76.6	39.8
MFPT [58]	RGB+IR	80.0	41.2
LRAF-Net [59]	RGB+IR	80.5	**42.6**
ICAFusion [43]	RGB+IR	79.2	41.2
ProbEn3 [60]	RGB+IR	80.1	41.9
RSDet [61]	RGB+IR	**81.1**	41.4
CrossFormer [62]	RGB+IR	79.3	42.1
CFIMNet (Ours)	RGB+IR	83.4	42.8

large-scale variation and strong interclass similarity in aerial imagery.

On the FLIR dataset, the comparison results are reported in Table VII. CFIMNet achieves the best overall performance, reaching 83.4 mAP50 and 42.8 mAP. Compared with the existing multimodal detection methods, CFIMNet shows a clear advantage in terms of mAP50, while also obtaining the highest mAP, which demonstrates its strong overall detection capability on this challenging benchmark. In particular, even when compared with representative multimodal methods, such as LRAF-Net, RSDet, and CrossFormer, CFIMNet still achieves more competitive results, indicating that the proposed framework can better exploit the complementary information of RGB and infrared modalities. Since FLIR contains abundant day-night scene transitions and complex thermal conditions, this performance improvement suggests that the proposed CFE and iterative multiscale fusion strategy can effectively strengthen structural representation and cross-modal semantic consistency, thereby improving detection robustness under challenging day–night and thermal imaging variations.

2) Qualitative Evaluation: To further provide an intuitive comparison of detection quality, we visualize the detection results of CFIMNet and several representative methods on the DroneVehicle and VEDAI datasets, as shown in Figs. 6 and 7, respectively. In the DroneVehicle examples, CFIMNet produces more complete detections with fewer false-positive detections and missed detections than C^{2}Former, ICAFusion, SuperYOLO, and FFCA-YOLO. In particular, for night scenes and low-contrast targets, other methods still exhibit missed detections or inaccurate responses around small vehicles, whereas CFIMNet can more accurately localize object boundaries and produce more complete target detections, showing stronger robustness to low illumination and cluttered backgrounds.

Similarly, in the VEDAI visualization results, CFIMNet shows clearer advantages in scenes containing small, densely distributed, or background-confounded targets. As seen in Fig. 7, competing methods often produce either false alarms in surrounding structures or missed detections for weak and tiny objects, while CFIMNet yields cleaner prediction results with better suppression of background interference. This indicates that the proposed CFE mechanism can strengthen local texture and contour representation, and the iterative multiscale fusion

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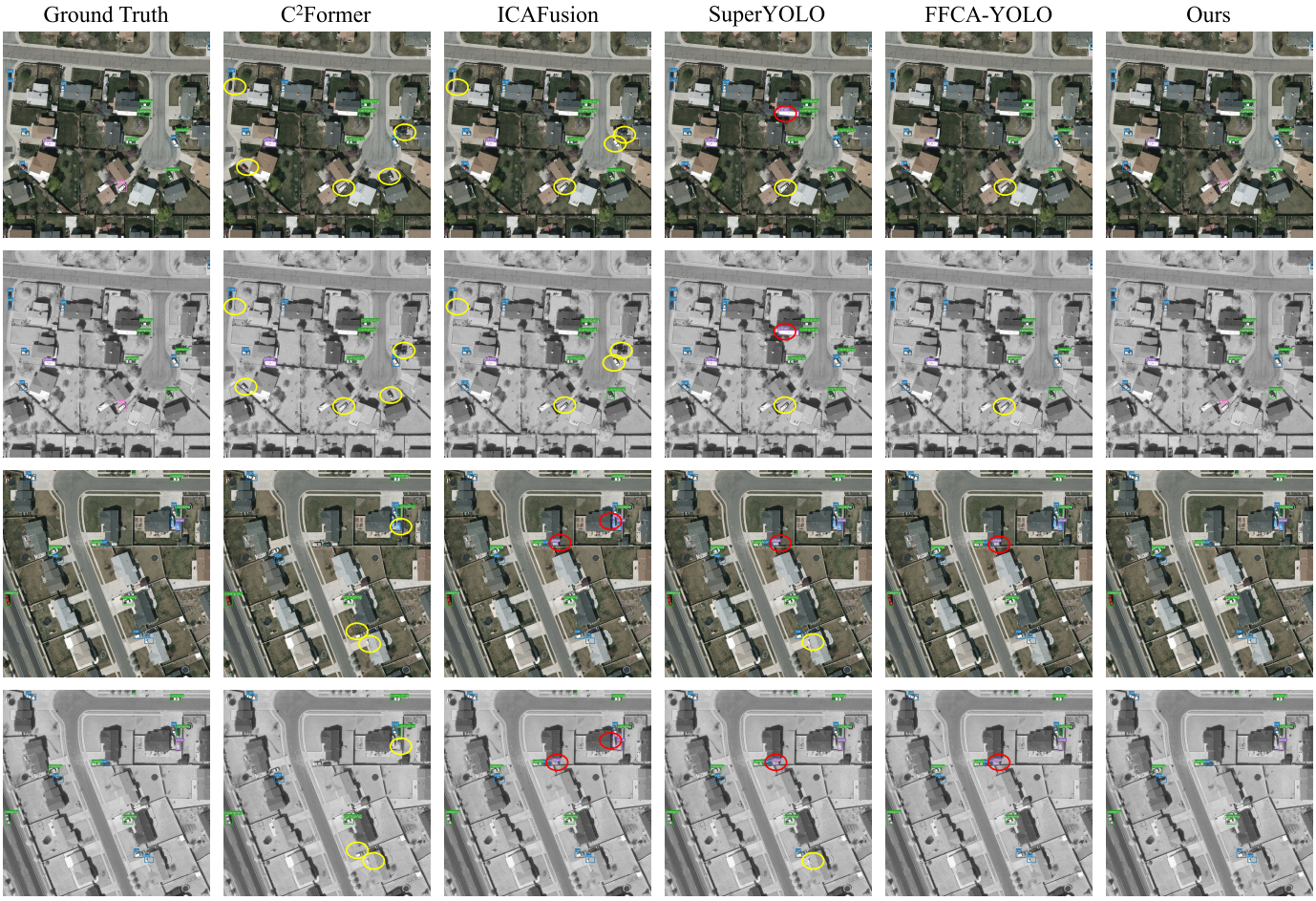


Fig. 7. Qualitative comparison on the VEDAI dataset. Red circles indicate false-positive detections and yellow circles indicate missed detections.

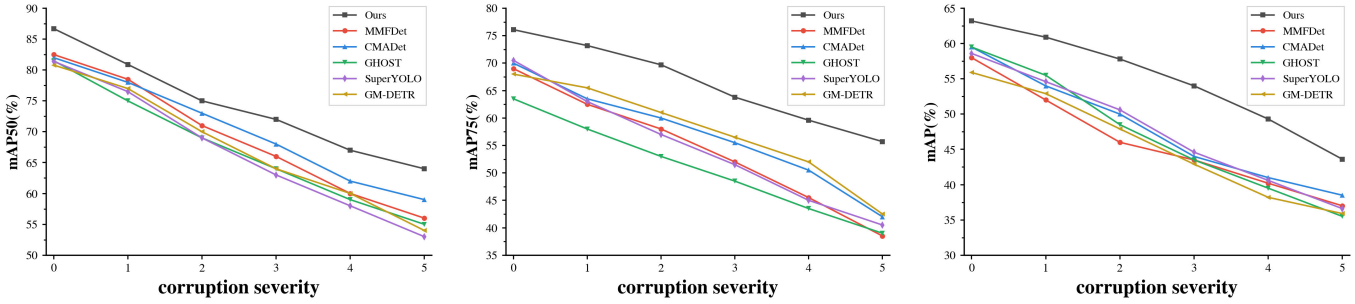


Fig. 8. Robustness comparison results with the state-of-the-art methods under image degradation on the DroneVehicle dataset.

module can progressively refine the fused features, leading to more stable detection in complex aerial scenes. Overall, the visual comparisons are consistent with the quantitative results and further verify that CFIMNet has stronger robustness and discriminative capability for multimodal remote sensing object detection.

3) Robustness Comparison Under Image Degradation: Using the same corruption protocol as in the ablation experiments, we further compare CFIMNet with representative state-of-the-art detectors under increasing degradation severity. As shown in Fig. 8, the performance of all methods decreases as the corruption severity increases, indicating that image degradation reduces target discriminability and cross-modal representation quality. Nevertheless, CFIMNet achieves the highest mAP50, mAP75, and mAP across all evaluated severity levels.

Compared with the competing methods, CFIMNet exhibits a consistently slower performance degradation as corruption severity increases. This advantage is particularly evident at moderate and high corruption levels, suggesting that the combination of frequency-aware cross-modal enhancement and iterative multiscale refinement improves robustness under degraded observations.

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(a) (b) (c)

Fig. 9. Failure cases on the DroneVehicle, VEDAI, and FLIR datasets. (a)–(c) Examples from DroneVehicle, VEDAI, and FLIR, respectively. Yellow circles indicate missed detections.

E. Limitations

Although CFIMNet achieves the best overall performance among the compared methods on the three benchmark datasets, failure cases remain under challenging imaging conditions, as illustrated in Fig. 9. The yellow circles mark missed detections. In dark or low-contrast aerial scenes, small vehicles may be difficult to distinguish from the surrounding background when both RGB appearance cues and infrared responses are weak. Targets adjacent to building edges, shadows, road markings, or other structured backgrounds may also be suppressed because their local appearance and thermal responses are insufficiently distinctive. Similar failures occur in ground-level multispectral scenes, where distant pedestrians and small objects may be missed because of low spatial resolution, occlusion, illumination interference, or background clutter.

These failure cases suggest that, although the proposed CFE and iterative multiscale fusion strategies improve feature complementarity and robustness, they may still be limited when target responses are extremely weak or when background regions exhibit target-like structures. In future work, we will further explore hard-sample mining, uncertainty-aware fusion, and fine-grained small-object enhancement strategies to improve detection reliability under conditions involving severely reduced visibility, occlusion, and cluttered backgrounds.

V. Conclusion

In this article, we proposed CFIMNet, a cross-modal frequency-guided IMFF network for RGB–infrared remote sensing object detection. CFIMNet comprises two complementary modules. The CFE module performs channel–spatial interaction and adaptive frequency decomposition to strengthen structural information and fine-grained details in modality-specific features. The IMFF module progressively refines RGB and infrared representations across spatial scales using shared parameters, thereby improving cross-scale semantic consistency while preserving modality-specific characteristics. Experiments on the DroneVehicle, VEDAI, and FLIR datasets demonstrate that CFIMNet consistently outperforms the compared methods in terms of detection accuracy and robustness. Despite these improvements, frequency-domain processing and iterative refinement introduce additional computational overhead. The method may also remain sensitive to severe cross-modal misalignment, simultaneous degradation of both modalities, and extremely weak small-object responses. Future work will focus on lightweight implementation, alignment-aware fusion, and more robust cross-modal representation learning for large-scale remote sensing applications.

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